

MATH0128 Harmonic Analysis

<i>Year:</i>	2026–2027
<i>Code:</i>	MATH0128
<i>Level:</i>	7 (UG/PG)
<i>Normal student group(s):</i>	UG: Year 4 Mathematics degrees PG: MSc Mathematics
<i>Value:</i>	15 credits (= 7.5 ECTS credits)
<i>Term:</i>	2
<i>Assessment:</i>	90% examination, 10% coursework
<i>Normal Pre-requisites:</i>	MATH0017 Measure Theory
<i>Lecturer:</i>	Prof C Marinelli

Course Description and Objectives

The module introduces the basics of harmonic analysis, with emphasis on topics that find applications in other fields of mathematics, in particular the analysis of partial differential equations. Students will be able to work with the standard basic methods of harmonic analysis for real-valued functions on Euclidean spaces.

Recommended Texts

J. Duoandikoetxea, *Fourier analysis*, American Mathematical Society, Providence, RI, 2001.

N. V. Krylov, *Lectures on elliptic and parabolic equations in Sobolev spaces*, American Mathematical Society, Providence, RI, 2008.

Detailed Syllabus

Review of measure theory.

L^p and weak L^p spaces, interpolation theorems of Riesz-Thorin and Marcinkiewicz.

Partitions, stopping times, Calderon-Zygmund decomposition. Maximal and sharp functions, Fefferman-Stein theorem. Sketch of applications to the L^p theory of second-order elliptic PDEs.